NASA SPACE APPS CHALLENGE 2026 // BE AN EARTH SYSTEM TREND DETECTIVE!

THE MILAN HEAT CASE: MAPPING CLIMATE INEQUALITY

LIVE APPLICATION: milan-heat-explorer.streamlit.app	**3-MIN VIDEO BRIEFING:** youtu.be/7MqjsmNJB54
SOURCE REPOSITORY: github.com/marybilik/milan-heat-explorer	**STUDY DOMAIN:** Milan, Italy (6,079 Census Tracts // 2000–2023)

CENTRAL EARTH SYSTEM DETECTIVE QUESTION & THESIS

Question: Are socially vulnerable communities in Milan experiencing a statistically significant acceleration in surface warming compared to affluent neighborhoods, what physical and industrial mechanisms drive these local trends, and which interventions actually work?
Answer: Yes. Across 6,079 census tracts, Land Surface Temperature (LST) diverges by **6.87°C within 300 meters** (peaking at **54°C**). Over the past decade, vulnerable tracts (Q4) warmed at **+0.858°C/decade** vs. **+0.825°C/decade** in affluent tracts (Q1)—a widening gap confirmed statistically significant after spatial lag control (**SVI p < 0.0001**, compounding by **~0.33°C per decade**).

01 MOTIVATION & PROBLEM STATEMENT

Why Gamify Earth Observation Data? Satellite climate records and spatial regressions are high-dimensional, intimidating, and often locked inside static academic papers. Municipal policymakers, urban planners, and citizens rarely have time to parse econometrics. *Data tells, but stories sell.* By turning the user into an Earth System Trend Detective across five interactive investigation phases, our tool transforms raw NASA/ESA orbital telemetry into actionable civic intelligence.

The Urban Climate Inequality Problem: Standard urban heat policies rely on two flawed assumptions: (1) that heat risk is evenly distributed across concrete neighborhoods, and (2) that planting street trees everywhere solves urban warming. Our investigation of Milan exposes five structural realities hidden from conventional heat maps:

- **The Physics Illusion:** Impervious surfaces and lack of canopy explain **57.5%** of LST variance; social vulnerability (SVI) adds 0% incremental R^2 over physics, proving that historical urban planning sorted vulnerable populations into unshaded concrete corridors.
- **The Loneliness Trap:** After strict Benjamini-Hochberg False Discovery Rate (FDR) correction across eight demographic variables, traditional scapegoats (such as foreign nationals, $p = 0.14$) are rejected; **one-person households** emerge as the strongest demographic heat predictor (**+0.083°C per SD, FDR p < 0.0001**).

- **The Industrial Ovens: 59.2% of extreme heat anomalies** (253 tracts > +2°C above physical model predictions) intersect industrial, logistics, and rail infrastructure—a **4.3× concentration** over the 13.9% city baseline.
- **The Invisible Majority:** Bivariate LISA clustering identifies **589 Red tracts** (High Heat + High Vulnerability) alongside **687 Orange tracts** (Low Heat + High Vulnerability). Because **49.3% of poor residents** live in "Mild/Warm" zones, red-zone-only heat plans miss half of the at-risk population.
- **The Tree Saturation Ceiling: 64.3% of poor tracts** have already reached the physical tree-cover metric ceiling (0.614), rendering marginal tree planting ineffective (**+0.04°C**) compared to **Cool Reflective Roofs (-0.36°C)**.

02 MISSION TELEMETRY: KEY QUANTITATIVE FINDINGS

Mission Metric	Measured Value	Empirical & Operational Context
Total Population in Risk Zones	**571,146** (40.8%)	2,222 tracts with 1 risk factor (~511,728 residents); 258 tracts with 2+ overlapping risks (~59,417 residents).
Micro-Scale Thermal Contrast	**6.87°C** in 300 m	Summer 2023 Landsat 8/9 LST contrast between historic city core (Duomo) and Parco Sempione.
Extreme Heat Exposure (>50°C LST)	**19.2%** vs. **2.5%**	Vulnerable tracts (Q4) are **7.6× more likely** to exceed 50°C surface temperature than affluent tracts (Q1).
Seasonal Amplification (Q4 − Q1 Gap)	**+2.35°C** Summer	Shifts from **-0.15°C** (Winter) to **+1.28°C** (Spring), peaking at **+2.35°C** (Summer) and **+1.50°C** (Autumn).
Decadal Heating Rate (2014–2023)	**+0.858°C** vs. **+0.825°C**	Q4 warms +0.033°C/yr faster than Q1; Spatial Lag Model confirms significance (Moran's I = 0.74, **p < 0.0001**).
Extreme Tail Risk Escalation	**10×** at Q95	Quantile regression shows SVI impact escalates sharply in extreme heat (Q75: +0.0343°C; Q95: +0.0791°C).

03 ORBITAL & GROUND DATA ARCHITECTURE (RESOURCE SELECTION)

In strict compliance with NASA Open Data rules, we selected multi-sensor satellite archives on **Google Earth Engine** to balance fine-grained spatial resolution (30 m) against multi-decadal temporal depth (24 years), integrated with census vulnerability and 3D ground-truth imagery:

Sensor / Dataset	Catalog ID / Source	Resolution	Selection Rationale & Analytical Role
NASA/USGS Landsat 8 & 9 C2 L2	`LANDSAT/LC08/C02/T1_L2` `LANDSAT/LC09/C02/T1_L2`	30 m (2014–2023)	Captures street-level thermal heterogeneity across 6,079 small urban census tracts, seasonal contrasts, and 10-year tract slopes.
NASA LP DAAC MODIS LST & NDVI	`MODIS/061/MOD11A2` `MODIS/061/MOD13A2`	1 km (2000–2023)	Provides an uninterrupted 24-year climate baseline to test long-term Q4 vs. Q1 warming and vegetation divergence.

Sensor / Dataset	Catalog ID / Source	Resolution	Selection Rationale & Analytical Role
ESA Copernicus Sentinel-5P	`COPERNICUS/S5P/OFFL/L3_NO2`	~5.5 × 3.5 km (Summer 2023)	Quantifies tropospheric NO_2 air pollution for the "Double Burden" index (cloud-missing pixels explicitly dropped, never zero-filled).
EC JRC GHSL Population	`JRC/GHSL/P2023A/GHS_POP/2020`	100 m (2020)	Replaces uniform tract estimates with true population-weighted exposure (identifying non-residential parks/airports with 0 residents).
Milan Social Vulnerability Index	Zenodo Records `15675828` & `13894260`	6,079 Tracts (2021 Census)	Supplies composite SVI plus 8 demographic sub-indicators (age, isolation, unemployment, education) and physical cover metrics.
OpenStreetMap & Google Earth 3D	OSM (`osmnx`) & Google Earth 3D Basemaps	Vector Polygons & Sub-meter 3D	Validates whether positive regression residuals (> +2°C) correspond to industrial, logistics, and railway infrastructure.

04 VISUAL GROUND-TRUTH: EXPOSING THE "INDUSTRIAL OVENS"

Mapping residuals from our physical OLS model isolates **253 extreme anomaly tracts** running **5–7°C hotter** than concrete cover alone predicts. Spatial intersection with OSM industrial polygons shows **59.2% overlap** (vs. 13.9% city baseline). Overlaying these residual polygons onto 3D Google Earth imagery confirms that unshaded freight yards and rail corridors act as massive thermal batteries bordering dense residential blocks:



VISUAL EXHIBIT A // LOGISTICS & BUS DEPOTS

Freight Yards & Fleet Depots (https://lh3.googleusercontent.com/gg/AJWXcNfGGiFCOsrFzR7TogcyZ8Xn540pJmkS_QMlUOM5MK4HK38Jghn8iZZZ0ZwjzyCmTMOq5EaldUGoV64w8z7o11P8a-mzjDpTl-pu0-LLgclLy-363MlnGpAPTbQg85K0CT0PgLMDdOXdOoR2BCnc4BgmlCesMnh7gjOmoxRiTobelpd_j3lOooC3aq-V_Q1L26wEWFLoQewf1UrchvTub-Vt_BfToRUfgsejevFRQhZHC5GcM6G36S42yFNcKQ4y1thVjha6QNXlpmjmVsUAflkodX1u38_HoQ=s0)

Red anomaly overlay aligns directly with wide asphalt loading bays and bus depots, radiating intense stored heat into adjacent apartment blocks.



VISUAL EXHIBIT B // RAILWAY INFRASTRUCTURE

Triboniano / Certosa Rail Corridor (https://lh3.googleusercontent.com/gg/AJWXcNf_W-_3Dt-SaiLPAKECugrGXuXiBJncyEvKD_JH6YKeCFi4gcrvGtCJgdSCrcjRjLk1PPgDbV2GERwOJuLKm7sAWJbL2il0-7jQ9JkuyFDauR_d5Mcl6Ooy3Grdjhqisn7_g_CrtK0socqKv7zOYgMatvlae1wA1VAzSi_yBDqCUjlIlJt9fsU2FlGgpGkFv10w4ASlBqyC-kfJ87Ax-51va-2T7GJlrNnHEFqq0hfF1Ko5RUffW5dJGh9HDMSPT_lx564j1IvA8-2M5Pru3PH37LHHtJiKzf0=s0)

Unshaded steel rails, stone ballast beds, and industrial warehouses generate a continuous localized heat island immediately bordering residential housing.

05 END-TO-END SCIENTIFIC METHODOLOGY & VALIDATION

To meet the *Advanced* rigor of the *Trend Detective* challenge, every spatial and temporal finding was verified against multicollinearity, spatial autocorrelation, spatial leakage, and extreme-tail nonlinearity:

Analytical Stage	Statistical / ML Method	Key Diagnostic & Empirical Outcome
1. Variance Decomposition	Nested OLS Regressions + Variance Inflation Factor (VIF)	Physics alone yields **R^2 = 0.575**; adding SVI yields **R^2 = 0.575** (full physical mediation).
2. Demographic Disaggregation	8 Controlled Regressions + Benjamini-Hochberg FDR	Isolates **one-person households** (+0.083°C/SD, **FDR p < 0.0001**); rejects false positives.
3. Trend & Spatial Lag Control	Mann-Kendall, Sen's Slope (95% CI) & ML Spatial Lag Model	Corrects for Moran's I = 0.74; SVI heating trend remains highly significant (**p < 0.0001**).
4. Spatial Clustering & GWR	Univariate/Bivariate LISA & Geographically Weighted Regression	Identifies **589 Red** & **687 Orange** tracts; GWR local SVI slopes span **-1.9°C to +1.8°C**.
5. Tail Risk & Compound Index	Quantile Regression (Q50–Q95) & Weighted Rank Composite	SVI effect jumps **10× at Q95**; composite index maps North Milan LST + NO_2 double burden.

Analytical Stage	Statistical / ML Method	Key Diagnostic & Empirical Outcome
6. Spatial Validation & Policy Sim	5×5 Spatial Block CV (GroupKFold) & 300-Tree Random Forest	Honest Spatial CV **MAPE = 2.07%** (vs. 2.01% random split); Cool Roofs (-0.36°C) beat Trees (+0.04°C).

06 INTERACTIVE GAME ARCHITECTURE & MUNICIPAL ACTION PLAN

5-Phase Streamlit Investigation (`app.py`): Users solve the Milan heat mystery step-by-step: *Phase 1: The Crime Scene* (300 m Duomo vs. Park gap), *Phase 2: The Time Bomb* (seasonal amplification), *Phase 3: The Invisible Enemy* (OSM industrial anomaly toggle), *Phase 4: The Hotspots* (local GWR coefficient explorer), and *Phase 5: The Invisible Majority* (Bivariate LISA orange-zone discovery).

1. **Industrial & Rail Buffer Mandates (253 Anomaly Tracts, e.g., Top Target Tract 151460001206 at 53.4°C):** Require vegetative buffer strips, high-albedo warehouse roofs, and industrial heat-emission caps along logistics hubs and railway corridors.
2. **Cool Roof Subsidies Over Street Planting (Q4 Space-Saturated Tracts):** Redirect tree-planting budgets in dense tracts (where 64.3% hit the tree-cover ceiling) toward reflective cool roofs (**-0.36°C** simulated cooling) and permeable pavements.
3. **"Check Your Neighbor" Heat Registry (687 Orange Tracts & Isolated Seniors):** Deploy proactive medical outreach, cooling-center transit vouchers, and AC subsidies for one-person households and elderly residents before summer heatwaves peak.